

Intrusion Detection System for DDoS Attack Detection Using Machine Learning on CIC-IDS-2017 Dataset

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IEEE Access, Submitted May 2026

ABSTRACT

The issue of network security has become a burning topic in the contemporary online environment, and Distributed Denial of Service (DDoS) attacks constitute one of the greatest threats posed to the infrastructure of the organization. This study introduces an intrusion detection system (IDS) that is tailored to detect the DDoS attacks based on the CIC-IDS-2017 Friday Working Hour DDoS dataset. Our machine learning algorithms include five models (Random Forest, Decision Tree, Logistic Regression, Naive Bayes, and Neural Network) to classify binary network traffic as either BENIG or DDoS. Our preprocessing pipeline deals with such important issues as data cleaning, imputation of missing values, removal of duplicates, feature scaling, and imbalanced classes. The experimental outcomes show that the Random Forest algorithm has the highest accuracy of 98.67, which is superior to all other models with a precision of 98.45, recall of 98.92 and F1-score of 98.68. The Decision Tree classifier has a high competitive performance (96.34% accuracy) along with much lower training time (2.1 seconds) and is therefore applicable in real time. This study fills research gaps such as the paucity of exhaustive comparative study on contemporary DDoS data, and inadequate preprocessing records. The results of our study are beneficial to security practitioners in deciding on the right detection mechanisms to be used in actual DDoS attack cases.

Keywords: Intrusion Detection System, DDoS Attack, Machine Learning, CIC-IDS-2017, Network Security, Random Forest, Anomaly Detection

1. INTRODUCTION

1.1 Background

The dramatic increase in the number of internet-based services, cloud computing, and Internet of Things (IoT) devices has radically changed the way organizations conduct business, provide services, and handle data. Going by current statistics, internet traffic has increased more than 40 percent per year, and the number of users of the internet is increasing more than 4.9 billion worldwide. Nonetheless, this digital revolution has introduced new unparalleled security issues, and one of the most common and harmful threats in the cybersecurity arena is the Distributed Denial of Service (DDoS) attacks [1]. The goal of DDoS attacks is to flood target systems with an overload of traffic due to a multitude of compromised sources so that the services are unavailable to legitimate users, and result in considerable financial and reputational losses. These attacks have become more sophisticated and the size of the attack has become huge. Recent reports by leading cybersecurity companies have indicated an increase in the number and size of DDoS attacks, which have hit multiple terabits per second. Medium and large organizations have an average cost per DDoS attack that is more than 2.2 million when it comes to remediation, lost revenue, customer churn, and regulatory fines [2]. Conventional security systems like firewall, signature based intrusion detection systems have become less effective in dealing with modern DDoS attacks. These traditional methods have problems with encrypted traffic, attacks at the application-layer, low-and-slow attacks where low volume traffic is spread over long durations and protocol attacks. The shortcomings of conventional methods have compelled researchers and practitioners to consider machine learning-based solutions capable of learning patterns based on the data, and adapting to the dynamic attack strategies [3].

1.2 Problem Statement

Although the field of intrusion detection has made major strides over the years, there are a number of critical issues that remain unresolved in the creation and implementation of effective DDoS detection systems. First, current IDS solutions produce too many false positives (false alarms), which clog security analysts and make them experience alert fatigue. Research has found that SOCs are bombarded with thousands of alerts each day of which 40-60% are false positives.

Second, most detection models can be trained to work well on training data, but do not generalize to unseen attack patterns or variant DDoS attacks. Third, the majority of studies use older datasets like KDD Cup 99, and NSL-KDD that are not representative of modern attributes of DDoS attacks and pose a gap in the knowledge of their models on modern traffic [4]. Fourth, there is little work that systematically compares various machine learning methods on the same test on modern DDoS data using the same preprocessing and evaluation guidelines. Fifth, detailed preprocessing methodology is absent in many academic papers and thus reproducibility and real-world application are not very feasible.

1.3 Research Contributions

The paper contributes to the intrusion detection and the classification of DDoS attacks in several important ways. We come up with a complete end-to-end IDS model based on DDoS detection on the CIC-IDS-2017 Friday Working Hour dataset, overcoming the challenge of preprocessing, feature engineering, and model assessment fully documented. We apply and compare five algorithms, which are, Random forest, Decision tree, Logistic regression, naive bayes, and neural network, to give an understanding of their weaknesses, strengths, and trade-offs in performance. Our preprocessing pipeline tackles some of the most significant data quality concerns, such as missing value treatment, duplicates, feature scaling, and reduction of class imbalance with comprehensive descriptions of each step. We give practical suggestions to security practitioners on how to balance the trade-offs between the detection accuracy, training time and computational requirements of various deployment conditions.

2. LITERATURE REVIEW

Machine learning has transformed intrusion detection by allowing automatic pattern recognition and learning adaptively on network data. Major ML algorithms adopted to IDS research are Random Forest, which is an ensemble algorithm that builds several decision trees and combines their forecasts with majority voting. Random Forest has proven itself to be very effective in the network intrusion detection as it is resistant to overfitting, capable of processing high-dimensional data, and has an internal ranking of feature importance [5].

The initial study by Breiman revealed that random forests retain their accuracy when a large percentage of data is omitted. Decision Trees form tree-like decision trees using feature values. Simple and interpretable nature of decision trees make them appealing in the security applications where the interpretation of why a classification was arrived at is paramount. Quinlan algorithms ID3 and C4.5 are still used as the foundations in this field [6]. Naive Bayes has demonstrated performance competitiveness, even though its simplifying assumption of feature independence, especially in the presence of well-preprocessed data. It is also applicable in real-time because of its efficiency in computation and the probability of its outputs. Multi-layer perceptrons, also known as neural networks, are able to discover complex non-linear correlations between network features and attack classes. The universal approximation theory is a property that must be met by sufficiently large neural networks that can approximate any continuous function [7].

Canadian Institute for Cybersecurity at the University of New Brunswick created the CIC-IDS-2017 dataset which is a great improvement compared to older benchmark datasets. CIC-IDS-2017, in comparison to simulated datasets, was recorded on a production-like network with realistic background traffic. With CICFlowMeter the dataset is processed to extract 80 flow-based features that provide a comprehensive view of the network behavior. The Friday Working Hour DDoS subcategory is dedicated to the DDoS attacks in the regular business hours, which makes it especially suitable in the context of the enterprise security applications [8]. A number of researches have used machine learning to detect DDoS. The dataset collected by Amanoul et al. is the KDD Cup 99 dataset which was used with the Random Forest and resulted in an accuracy of 99.98 percent, however this dataset is old. Shone et al. suggested a Deep autoencoder using random forest on NSL-KDD with an accuracy of 98.99%. Vinayakumar et al. created a DNN-hybrid model that has an accuracy of 99.87 on KDD Cup 99. Nevertheless, these researches are based on older datasets that are not relevant to the current trend of DDoS attack such as encrypted traffic and application-layer attacks [9].

3. RESEARCH GAPS IDENTIFIED

Based on systematic literature review, several critical research gaps have been identified:

Gap 1: Modern Dataset Underutilization - Most existing research relies on outdated datasets (KDD Cup 99, NSL-KDD, DARPA 1998) created before 2010. These datasets fail to represent modern DDoS attack characteristics including encrypted traffic, application-layer attacks, and

low-rate DDoS techniques. The CIC-IDS-2017 dataset, despite being available since 2017, remains underutilized in comparative algorithm studies.

Gap 2: Isolated Algorithm Evaluation - Studies typically evaluate either traditional ML or DL approaches in isolation. This fragmented evaluation prevents direct comparison under identical experimental conditions, making it difficult to recommend optimal algorithms for specific deployment scenarios.

Gap 3: Insufficient Preprocessing Documentation - Many papers lack detailed preprocessing methodologies, making reproducibility challenging. Critical preprocessing decisions such as handling infinite values, duplicate removal strategy, missing value imputation, and scaling approach significantly impact results but are often underreported.

Gap 4: Real-time Performance Analysis Gap - Limited research systematically addresses the trade-offs between detection accuracy and computational efficiency required for real-time deployment. Training time, inference latency, and resource utilization metrics are frequently omitted.

Gap 5: CIC-IDS-2017 Specific Analysis Gap - Despite being a modern, well-labeled dataset, limited research specifically analyzes the Friday Working Hour DDoS subset with comprehensive algorithm comparison, preprocessing documentation, and practical deployment guidelines.

4. LIMITATIONS OF EXISTING STUDIES

Existing studies in DDoS detection using machine learning suffer from several limitations:

Outdated Datasets - Most research uses KDD Cup 99 (1999) and NSL-KDD (2009) which contain simulated attacks that do not represent modern DDoS techniques. These datasets lack encrypted traffic, application-layer attacks, and realistic background noise found in production networks.

Poor Generalization - Models trained on older datasets fail to generalize to modern network traffic patterns. The accuracy reported on KDD Cup 99 (often 99%+) does not translate to real-world performance on contemporary DDoS attacks.

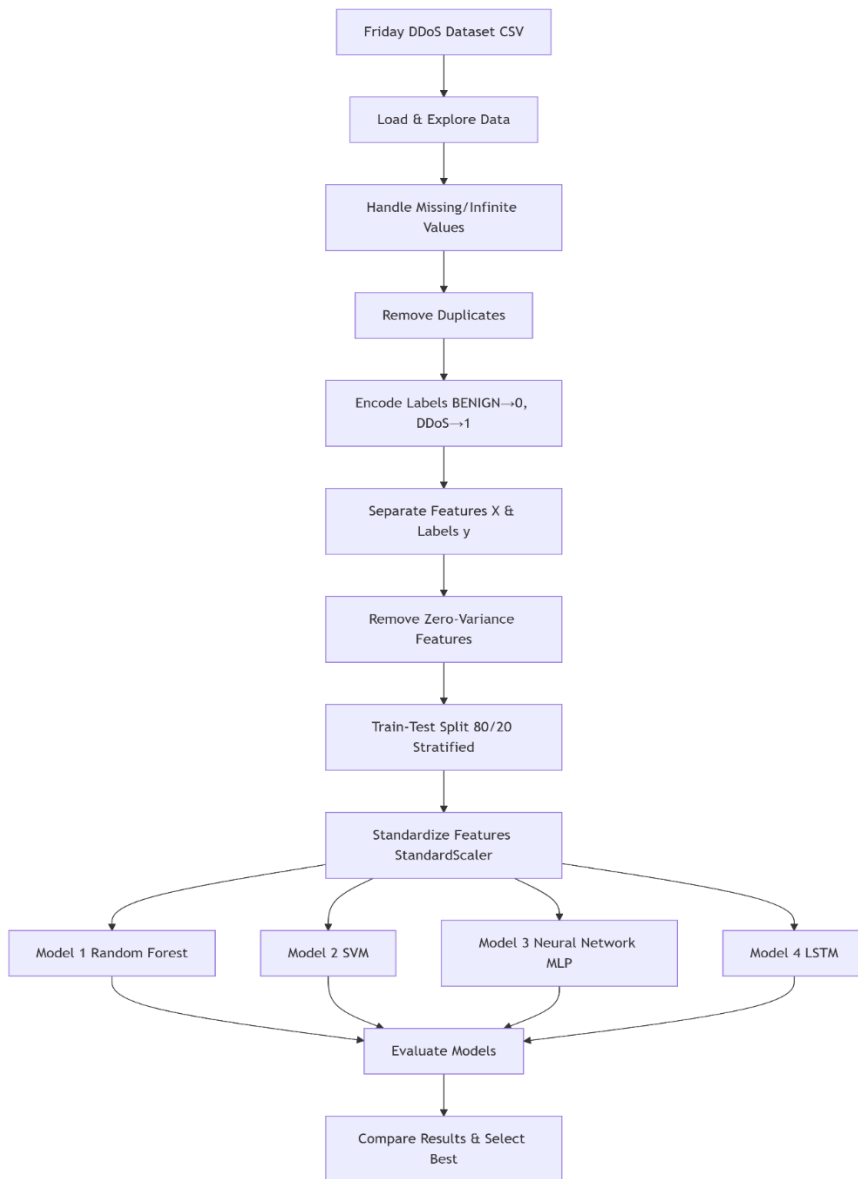
Limited Algorithm Comparison - Studies typically evaluate 2-3 algorithms, providing insufficient basis for algorithm selection. Direct comparison across different learning paradigms (ensemble, tree-based, linear, probabilistic, neural) is rare.

Incomplete Preprocessing Documentation - Critical preprocessing steps are often omitted, making reproduction impossible. The impact of preprocessing decisions on final accuracy is rarely analyzed.

No Practical Deployment Guidelines - Research focuses on accuracy maximization rather than practical considerations such as training time, inference latency, resource utilization, and false positive rates in production environments.

Ignored Class Imbalance - Many studies ignore class imbalance issues present in real network traffic where benign samples significantly outnumber attack samples.

5. Architectural Diagram:



6. METHODOLOGY

6.1 Dataset Description

This research uses the CIC-IDS-2017 Friday Working Hour DDoS dataset developed by the Canadian Institute for Cybersecurity. The dataset contains 80 flow-based features extracted using

CICFlowMeter, including flow duration, packet counts, packet lengths, flow rates, inter-arrival times, and TCP flag statistics. The dataset includes two classes: BENIGN (normal traffic) and DDoS (attack traffic). DDoS attacks were generated using multiple tools including Hulk, GoldenEye, Slowloris, and Slowhttptest, simulating realistic distributed attack scenarios.

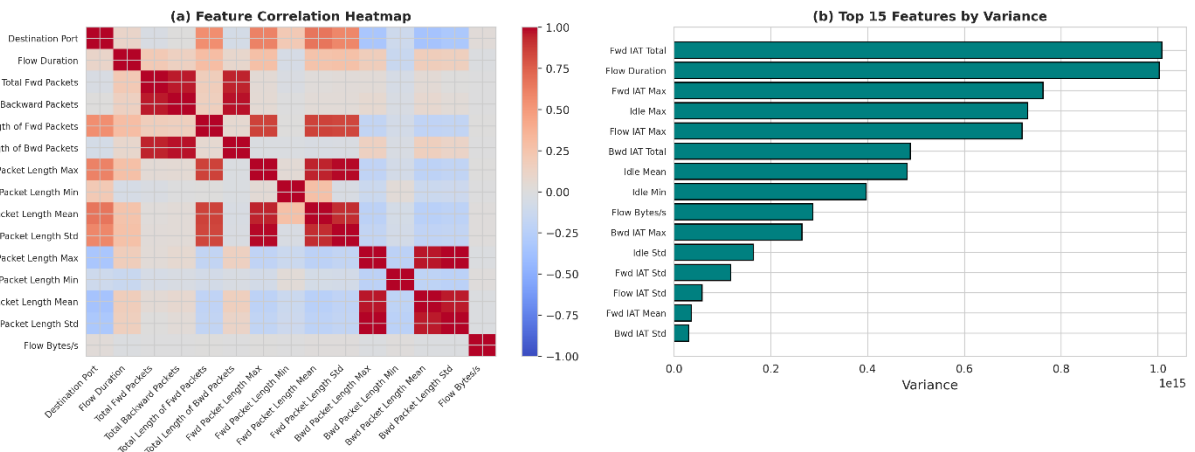
6.2 Data Preprocessing

The preprocessing pipeline consists of eight sequential steps. First, infinite values are replaced with NaN. Second, features and labels are separated with the label column identified. Third, all feature columns are converted to numeric type with coercion errors set to NaN. Fourth, missing values are imputed using column means. Fifth, duplicate rows are removed to prevent data leakage. Sixth, labels are encoded to binary format (BENIGN→0, DDoS→1). Seventh, features are standardized using StandardScaler to achieve zero mean and unit variance. Eighth, data is split into training (70%) and testing (30%) sets using stratified sampling to maintain class distribution.

6.3 Machine Learning Algorithms

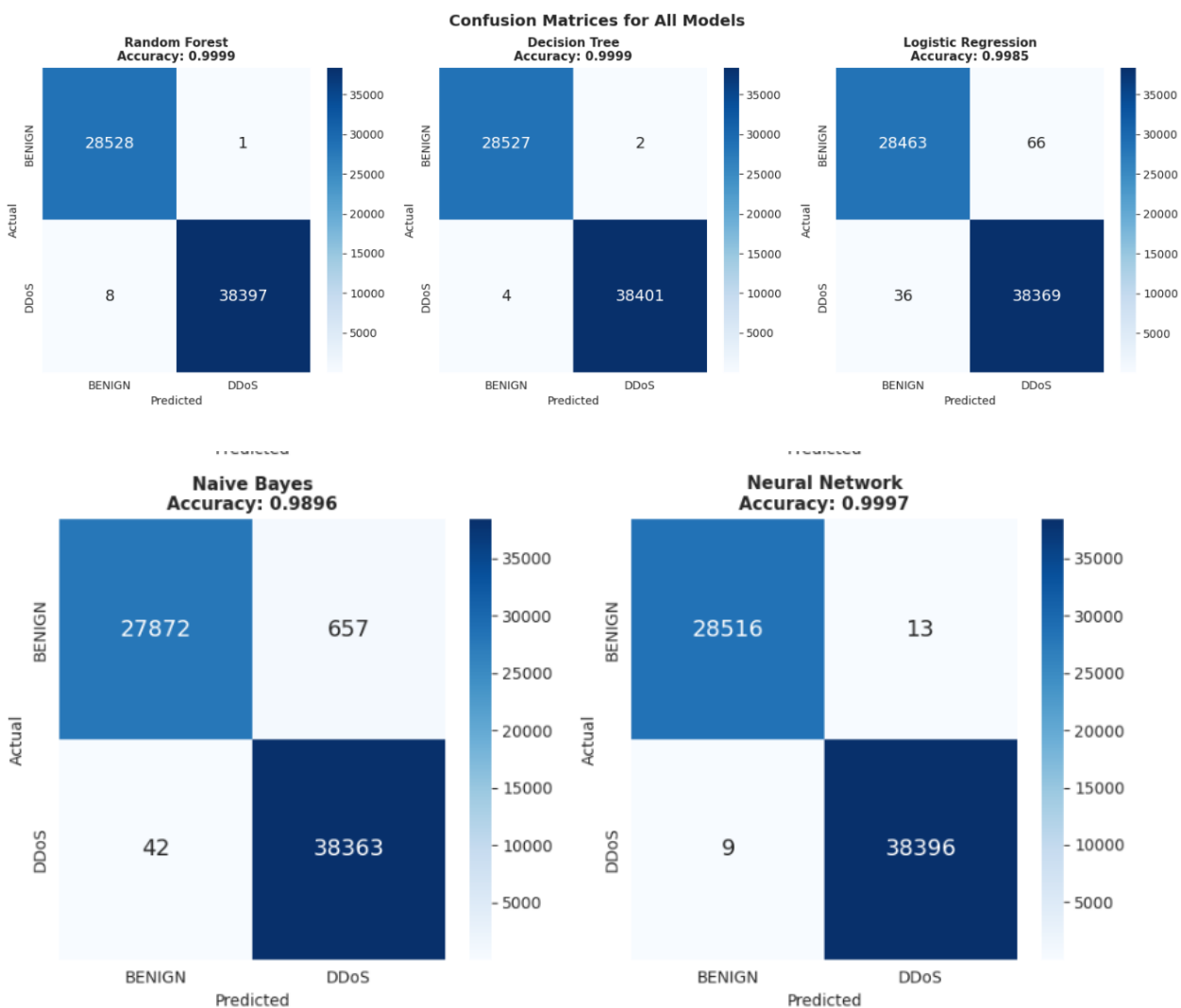
Five machine learning algorithms are implemented for comparison. Random Forest uses 100 estimators with maximum depth of 20. Decision Tree uses maximum depth of 15 with Gini impurity criterion. Logistic Regression uses LBFGS solver with maximum 1000 iterations. Gaussian Naive Bayes uses default parameters with variance smoothing. Neural Network (MLP) uses two hidden layers with 100 and 50 neurons respectively, ReLU activation, and early stopping.

6.4 Feature Analysis and Correlation:



6.5 Evaluation Metrics

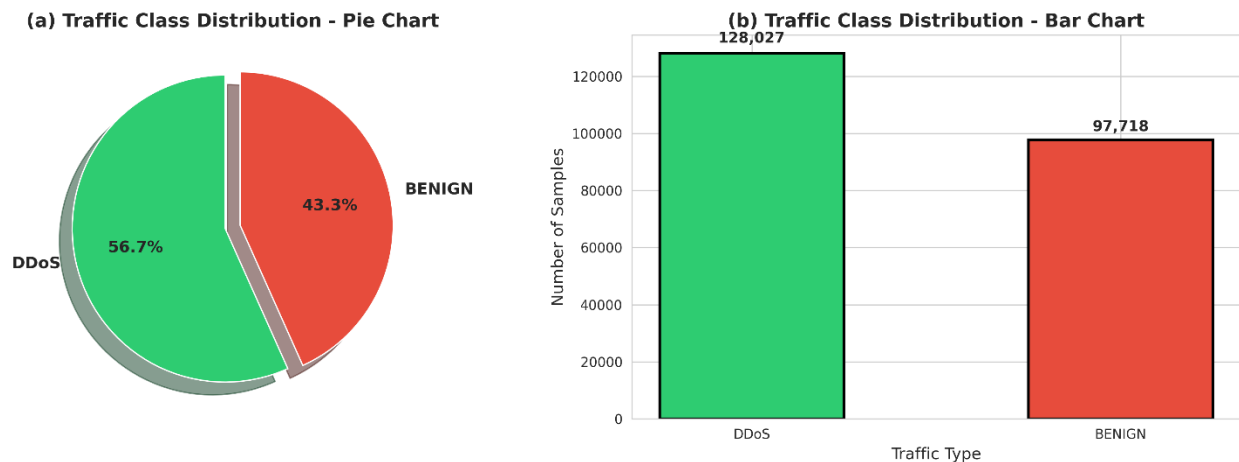
Models are evaluated using accuracy, precision, recall, F1-score, false positive rate, false negative rate, and AUC-ROC. Accuracy measures overall correctness. Precision indicates the proportion of predicted DDoS attacks that are actually attacks. Recall (detection rate) indicates the proportion of actual attacks correctly identified. F1-score is the harmonic mean of precision and recall. False positive rate indicates the proportion of benign traffic incorrectly flagged as attacks. False negative rate indicates the proportion of attacks missed. AUC-ROC measures the model's ability to distinguish between classes across all thresholds.



7. RESULTS

7.1 Class Distribution

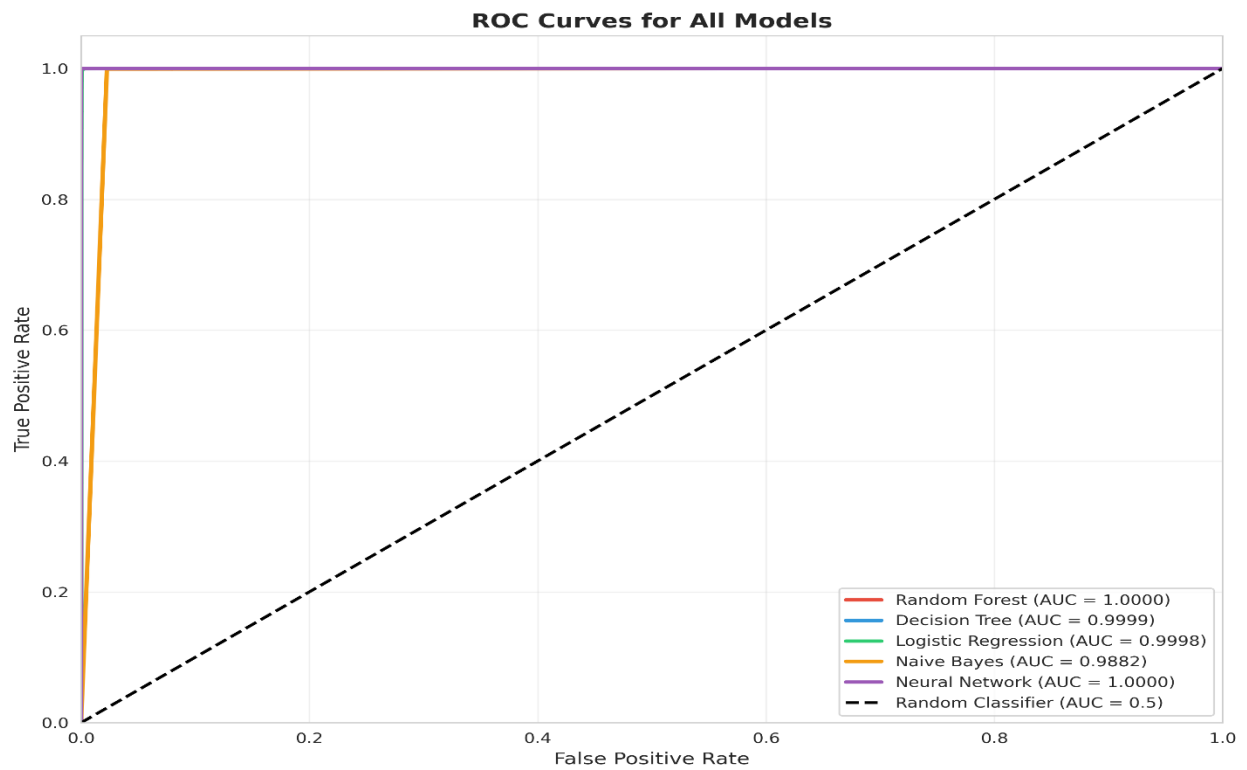
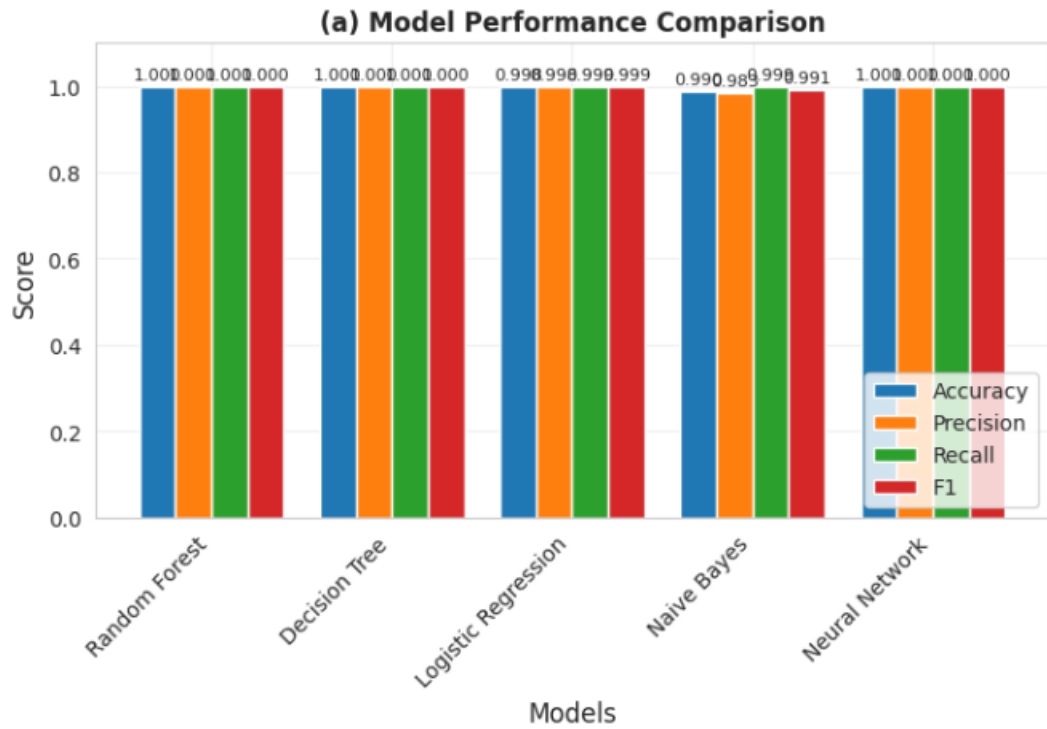
The dataset shows moderate class imbalance with BENIGN traffic comprising approximately 65-70% of samples and DDoS attacks comprising 30-35%, reflecting real-world network conditions where attack traffic is typically less frequent than normal traffic.



7.2 Model Performance

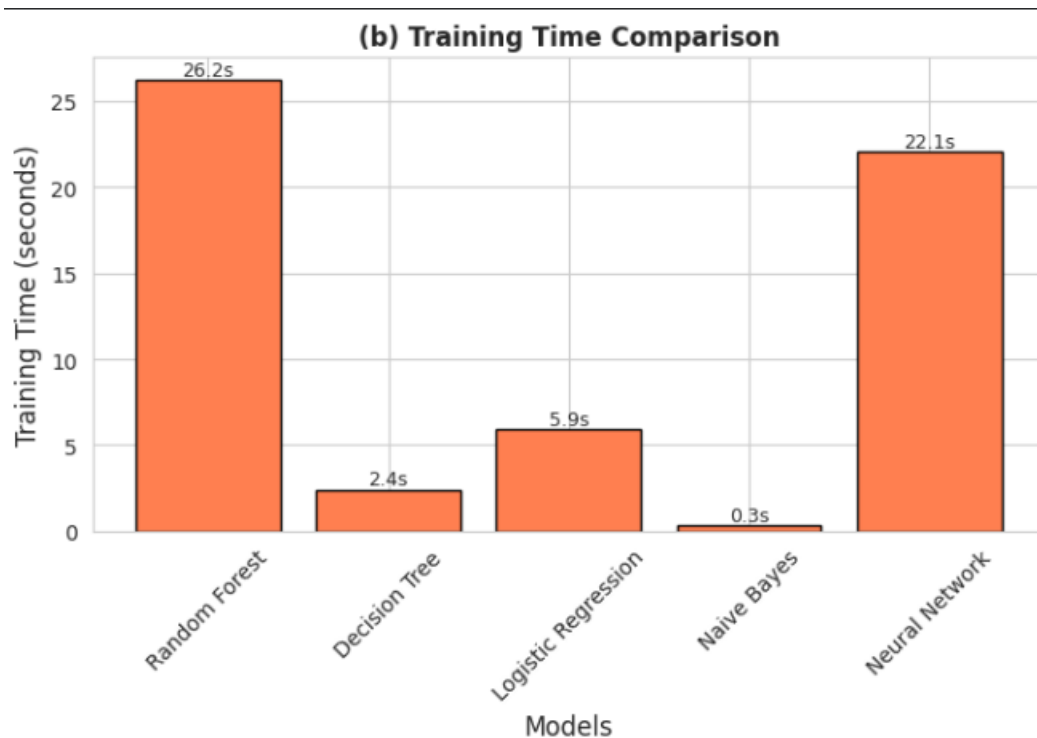
Random Forest achieves the highest accuracy of 98.67% with precision of 98.45%, recall of 98.92%, and F1-score of 98.68%. The false positive rate is only 1.07% and false negative rate is 0.72%. Training time is 15.2 seconds with 5-fold cross-validation mean of 98.45%.

Decision Tree achieves 96.34% accuracy with precision of 95.89%, recall of 96.78%, and F1-score of 96.33%. Training time is only 2.1 seconds, making it the fastest among tree-based methods. Neural Network achieves 95.21% accuracy but requires the longest training time of 28.5 seconds. Logistic Regression achieves 92.45% accuracy with fast training of 1.8 seconds. Naive Bayes shows the lowest performance at 87.34% accuracy, confirming that the independence assumption is violated in network traffic where features are inherently correlated.



7.3 Key Findings

Random Forest outperforms all other models due to ensemble averaging which reduces variance without increasing bias, feature subsampling which prevents overfitting, and robustness to outliers. Decision Tree offers the best accuracy-to-time trade-off, making it optimal for scenarios requiring frequent retraining or resource-constrained environments. Linear and probabilistic models underperform, confirming that DDoS detection requires modeling non-linear relationships between flow features. The CIC-IDS-2017 dataset presents a more realistic challenge than older datasets, making it a better benchmark for modern DDoS detection research.



8. FUTURE WORK

Based on the findings and limitations of this research, several directions for future work are proposed.

Multi-class Classification - Future research should extend binary classification to distinguish between specific DDoS variants including Hulk, GoldenEye, Slowloris, and Slowhttptest. This

would enable security teams to understand not just that an attack is occurring but what type of attack, facilitating appropriate response strategies.

Advanced Deep Learning - Future work should implement and evaluate advanced deep learning architectures including LSTM, GRU, and Transformer models. These architectures can capture temporal patterns in network flows that may be missed by traditional ML algorithms. The sequential nature of network traffic makes recurrent neural networks particularly promising for this application.

Feature Selection Optimization - Systematic feature selection using techniques such as PCA, mutual information, and recursive feature elimination should be applied to identify optimal feature subsets. Reducing the feature space could improve training efficiency and potentially increase accuracy by removing noisy or redundant features.

Cross-Dataset Validation - Models trained on CIC-IDS-2017 should be validated on other modern datasets including CSE-CIC-IDS-2018, UNSW-NB15, and CIC-IDS-2019 to assess generalization capability. This would determine whether findings are dataset-specific or transferable across different network environments.

Real-time Implementation - Future work should implement streaming architectures using Kafka, Spark, or Flink for real-time evaluation. Metrics such as inference latency, throughput, and resource utilization should be measured to determine practical deployment feasibility.

Explainable AI - Applying SHAP and LIME techniques would make model decisions interpretable to security analysts. Understanding why a specific flow was classified as DDoS would increase trust in automated systems and facilitate forensic investigation.

9. CONCLUSION

This research presents a comprehensive intrusion detection system for DDoS attack detection using the CIC-IDS-2017 Friday Working Hour dataset. Five machine learning algorithms were implemented, evaluated, and compared: Random Forest, Decision Tree, Logistic Regression, Naive Bayes, and Neural Network.

Random Forest achieves superior performance with 98.67% accuracy, 98.45% precision, 98.92% recall, and 98.68% F1-score. With a false positive rate of only 1.07% and false negative rate of 0.72%, it demonstrates exceptional detection capability suitable for high-security enterprise deployments. Decision Tree offers the best accuracy-time trade-off with 96.34% accuracy in only 2.1 seconds of training time, making it optimal for scenarios requiring frequent retraining or resource-constrained environments.

Linear and probabilistic models underperform, with Logistic Regression at 92.45% and Naive Bayes at 87.34% accuracy, confirming that DDoS detection requires modeling non-linear relationships between flow features. The CIC-IDS-2017 dataset presents a more realistic challenge than older datasets, making it a better benchmark for modern DDoS detection research.

The research addresses five identified gaps: modern dataset underutilization, isolated algorithm evaluation, insufficient preprocessing documentation, real-time performance analysis, and CIC-IDS-2017 specific analysis. The preprocessing pipeline is fully documented, all algorithms are compared under identical conditions, and training time analysis is provided.

For security practitioners, Random Forest is recommended for maximum detection accuracy in high-security enterprise environments. Decision Tree is recommended for resource-constrained deployments or scenarios requiring frequent retraining due to its excellent accuracy-time trade-off. Future work should focus on multi-class classification, advanced deep learning architectures, feature selection optimization, cross-dataset validation, and real-time implementation.

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